

ORIENTATION, NOT MAGNITUDE: THE CAUSAL STRUCTURE OF TASK-VECTOR INTERFERENCE IN MERGED LANGUAGE MODELS

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ABSTRACT

Model merging by task arithmetic works until it doesn’t, and the field diagnoses why with magnitudes: layerwise representation bias, deviations from cross-task linearity, parameter overlap. Tracking the exact layerwise cross-term of merged LLMs through a factorial ledger and intervening on it directly, we find that magnitude is insufficient—and inconsistent across model families—as a diagnostic axis. An exact decomposition of the layerwise flux shows it is dominated by *amplifying transport* of the existing cross-term ($\sim 65\text{--}70\%$ in both model families, gain >1 per late block), and erasing the term is undone by propagation—rebuilt to 99% of its norm at cosine 0.99—unless applied near the output; a basin test (six starting displacements, including $2\times$ overshoot and four structured wrong directions, all reconverging at cosine 0.83–0.99) establishes the carried direction as an attractor of the forward pass. That direction is causally load-bearing: erasure along it removes expressed interference dose-dependently and saturates at exact erasure, norm-matched wrong-direction controls fail or backfire, and instruction wrappers gate the effect—the same erasure finds $13\times$ less *relative* interference to remove under a wrapper that internally amplifies the cross-term (the wrapper drowns the interaction in a template-pinned main effect rather than shrinking it). Magnitude, by contrast, is at best a coarse correlate: float32 local generation tracks pair ordering only loosely, every magnitude metric we measure (local, cumulative, parameter-space, representation bias) points the *wrong way* across formats, and the striking $\pm 15\%$ “universality” of naive bf16 generation turns out to be quantization roughness—a cautionary result we document. Notably, task pairs whose local cross-term generation differs by at most $1.9\times$ differ by $14\times\text{--}337\times$ in causally removable interference. All 46 predictions were preregistered and frozen before their data; falsifications, including of our own headline expectations, are reported as such. Reduced replications at a second family and a larger scale reproduce the causal structure at all three model points. Format-wrapped evaluations, we conclude, read the gate—not the interference.

1 INTRODUCTION

Task arithmetic is disarmingly effective: add two finetuned weight displacements to a base model and, much of the time, both skills transfer [Ilharco et al., 2023]. When it fails, the failure is measured at the output—but *located* nowhere. A growing literature quantifies merging interference by the size of internal deviations: layerwise representation bias between merged and expert models [Yang et al., 2024], departures from cross-task linearity [Zhou et al., 2024], subspace overlap between task matrices [Gargiulo et al., 2025]. All of these are magnitude measurements, and all of them inherit the same untested assumption: **that where interference appears large is where interference is made, and that making less of it is what makes merges work.**

This paper tests that assumption with a factorial intervention and finds it wrong on both counts. We track the exact non-additive residue of merging—the layerwise cross-term $I_\ell = h_\ell^{AB} - h_\ell^{A0} - h_\ell^{0B} + h_\ell^{00}$ —through every block of a merged LLM, measure what each block *generates* versus what it *carries*, and then erase the carried term surgically to see what the output loses. The question, precisely:

*When two task vectors are added, where does their functional interference causally live—
in how much cross-term the network produces, or in what the network does with it?*

The answer replaces the magnitude picture in three steps (Figure 1). *First*, the layerwise flux of the cross-term is dominated by *transport*: an exact decomposition of every block’s contribution into transport of the existing term versus newly generated pieces attributes $\sim 70\%$ to transport, with per-block gain above 1 in the late network—and consistently, cumulative interference exceeds what any per-layer injection of the measured generation could supply ($2.6\times$ even the perfectly-aligned bound). Erasing the carried term confirms this dynamically: propagation rebuilds it to 99% of its

norm, at direction cosine 0.99, unless the erasure is applied too late to be undone—the carried direction is an **attractor** of the forward pass (basin-tested in Appendix G). *Second*, and causally: erasing along the cross-term’s own direction removes expressed interference dose-dependently and saturates at exactly full erasure, while every norm-matched structural control fails or backfires—and the same erasure that removes interference on raw prompts finds $13\times$ less to remove under an instruction wrapper, because the wrapper *gates the expression* of interference it internally amplifies. *Third*, magnitude—the axis the field measures—carries little of this: float32 local generation orders the task pairs only coarsely ($1.9\times$ against a causal contrast of $14\times\text{--}337\times$), every magnitude metric we measure points the *wrong way* across formats, and the seductive near-uniformity of naive bf16 generation ($\pm 15\%$ across all conditions) dissolves under a precision audit into weight-rounding roughness—a standalone methodological caution (§4). The paper’s central contrast: **task pairs whose local cross-term generation differs by at most $1.9\times$ differ by $14\times\text{--}337\times$ in causally removable interference.**

Our thesis, in one sentence: *in merged language models, interference is transported and amplified along a direction that propagation actively maintains and the prompt format gates at expression—magnitude, local or cumulative, is at best a coarse correlate and is inverted exactly where evaluations look.*

Contributions.

- **Transport dominates, and the direction is an attractor (§4, §5).** An exact per-layer decomposition $I_{\ell+1} = G_{\ell} + T_{\ell} + M_{\ell}$ (transport of the existing term vs. newly generated pieces) attributes $\sim 70\%$ of the flux to transport with late-network gain 1.08/block, on top of the arithmetic exclusion (cumulative cross-term at $2.6\times$ the coherent accumulation bound); a preregistered regrowth experiment shows erasure is undone by propagation— $\rho = 0.99$, direction cosine 0.99—except near the output, where the durable causal effect concentrates.
- **Magnitude is not the signal (§4, §7).** Float32 local generation orders task pairs only coarsely ($1.9\times$ vs. the $14\times\text{--}337\times$ causal contrast) and, like every magnitude metric we measure (cumulative norm, parameter cosine, SurgeryV2’s representation bias), is *larger* under instruction wrappers where expressed interference collapses $20\times$. Includes a cautionary methodological finding: the striking $\pm 15\%$ cross-condition uniformity of naive bf16 generation is 75–90% quantization roughness (float32 audit, §4.1).
- **Cross-term erasure (§5).** The first activation-level causal intervention on task-vector cross-terms: against four norm-matched structural controls and a five-point dose ladder—all predictions frozen before implementation and passed 3/3 seeds—erasure removes $\sim 19\%$ of expressed interference without damaging behavior (top-1 agreement 0.94–0.99), wrong directions backfire, and removable interference differs $14\times\text{--}337\times$ between pairs whose magnitudes agree to within 15%.
- **Expression gating (§6–7).** Instruction wrappers *amplify* internal cross-term generation (3/3 seeds, a preregistered prediction falsified in reverse) while collapsing its expression $20\times$; erasure confirms the gate causally ($13\times$ less removable interference when wrapped). Consequently, format-wrapped evaluations read the gate, not the interference—and no existing magnitude metric (CTL linearity, parameter cosine) sees this, though cumulative magnitude does coarsely predict held-out pair ordering (§7).

Collectively, these results relocate merging interference from “how much is produced” to “which direction survives propagation and reaches the output.” All hypotheses, thresholds, and controls were preregistered in a versioned freeze log; falsified predictions, including two of three headline expectations, are reported as such—the mechanism above is what survived. Preregistered reduced replications at a second family (Llama-3.2-1B) and a larger scale (Qwen2.5-7B) reproduce all three pillars (Appendices F, K); frozen follow-ups establish the attractor (§5) and place SurgeryV2’s bias with every other magnitude metric—coarse on pairs, blind to the gate (§7).

2 RELATED WORK

Task arithmetic and model merging. Task vectors compose skills by weight addition [Ilharco et al., 2023, Ortiz-Jimenez et al., 2023]; a large literature improves merging by resolving parameter-space conflicts—sign agreement [Yadav et al., 2023], layer selection [Chen et al., 2025], singular-vector overlap [Gargiulo et al., 2025], neuron-level decomposition [Fang et al., 2025]—or by post-hoc repair of representation bias [Yang et al., 2024]; recent variants minimize interference in task-vector input subspaces [Cheng et al., 2025], layerwise feature drift in closed form [Sun et al., 2025], or directional consistency of task-vector singular spaces [Zhang et al., 2026]. These methods *presuppose* an account of where interference lives; our results test and revise it. “Direction” here means three things that should not

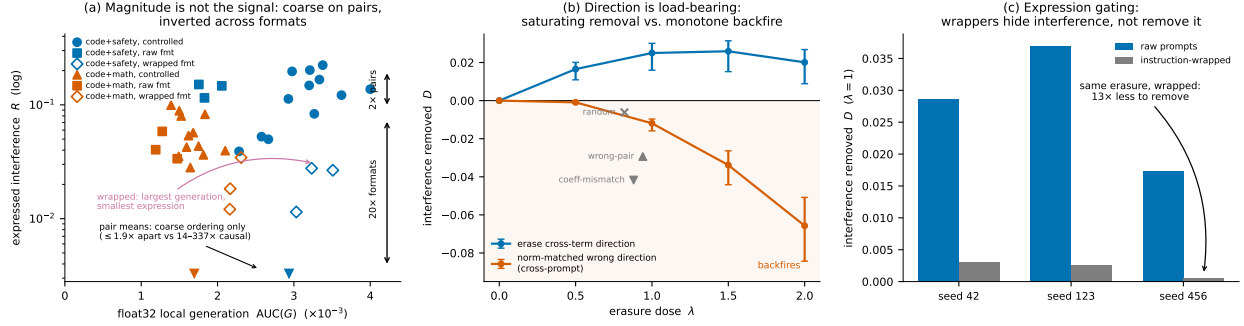


Figure 1: **Orientation, not magnitude: task pairs whose local cross-term generation differs by at most $1.9\times$ differ by $14\times$ – $337\times$ in causally removable interference (b), and instruction wrappers amplify internal interference while suppressing its expression 13-fold (a, c).** (a) Float32 local generation $AUC(G)$ (from the preregistered fp32 grid, Appendix L) versus expressed interference R , all 36 cells at the strongest composition point. Magnitude orders the pairs only coarsely (cluster means $\leq 1.9\times$ apart; triangles on the axis) and is *inverted* across formats: wrapped conditions (open diamonds) generate the most and express the least. (b) Dose–response of cross-term erasure: removing the cross-term along its own direction ($h_{\ell}^{AB} - \lambda I_{\ell}$) saturates near full erasure; the norm-matched cross-prompt direction backfires monotonically; grey markers are single-dose structural controls. Error bars: seed range. (c) The identical erasure removes $13\times$ less interference under an Alpaca instruction wrapper—the wrapper hides interference rather than removing it.

be conflated: *parameter-space* direction (the level of TSV and DC-Merge), *activation-space* direction (the orientation of I_{ℓ} —what we measure), and *causal* direction (what intervening on that orientation does—what we test); the third is load-bearing, while magnitude at any level is not a consistent signal. Activated Parameter Locating [Kong et al., 2024] applies causal intervention to merging at the *parameter* level; our intervention operates on *activations*, targeting the cross-term itself rather than parameter importance.

Linearity of features under weight interpolation. Cross-Task Linearity [CTL; Zhou et al., 2024] is the closest prior work: it observes that layerwise features of merged models are approximately linear combinations of the endpoint models’ features. Our I_{ℓ} is the residual of that approximation, so our measurements refine theirs: in float32 the approximation is even better than reported (linearity cosine 0.93–0.98 everywhere, §7). The difference is what each work concludes from the smallness. **CTL establishes that the layerwise cross-term is small; we establish that this small residual—specifically its orientation, not its magnitude—is where all of the functional interference causally lives, which magnitude-level analysis cannot see** (Table 1, §5–7). CTL contains no intervention, no format factor, and no propagation analysis.

Format sensitivity of merged models. RAIN-Merging [Huang et al., 2026] documents, at the behavioral level, that merging disrupts instruction-format handling and repairs it in the null space. Our expression-gating result is the mechanism-level counterpart—the wrapper closes an output gate on internally amplified interference—and neither result subsumes the other: they repair behavior; we locate what the behavior hides (§6).

Validity of activation patching. Our intervention inherits the general caveats of activation-level counterfactuals: patched states may interact with unpatched components [Vaidyanathan et al., 2026]. We therefore make no natural-mediation claims; every effect is reported as the response to a specified, norm-matched intervention, with each control’s directional overlap disclosed so that direction-mediation is checked rather than assumed (§3.4).

3 THE FACTORIAL LEDGER AND CROSS-TERM ERASURE

The intuition first. Merging two task vectors and running a prompt gives four natural models—base, each single task, and the merge—and the *interference* at any point in the computation is exactly what the merge has that the two single tasks do not add up to. Our method is nothing more than taking this four-way difference seriously at every layer, and then removing it surgically to see what changes.

Table 1: **No prior quantity is causal, directional, and format-aware; none explains the $2\times/20\times$ structure** (empirical evaluation in §7). Rows: CTL [Zhou et al., 2024], SurgeryV2 [Yang et al., 2024], TSV [Gargiulo et al., 2025], NeuroMerging [Fang et al., 2025], APL [Kong et al., 2024], RAIN-Merging [Huang et al., 2026]. Mag./Dir.: magnitude vs. direction quantity; Interv.: causal intervention on the quantity itself. CTL, parameter cosine, and SurgeryV2’s representation bias are *measured* in our conditions (§7, Appendix H), not only cited.

	Mag./Dir.	Interv.	Format	$2\times$ pairs	$20\times$ formats	Held-out
CTL	mag.	–	–	no	no	–
SurgeryV2	mag.	–	–	no	no	–
TSV	dir. (param.)	–	–	weak ($\rho=.25$)	no	weak
NeuroMerging	mag.	–	–	no	no	–
APL	mag.	param.	–	no	no	–
RAIN-Merging	(behav.)	repair	✓	–	behav.	–
Ours	dir. (activ.)	✓ (4 ctrl.)	✓	$14\text{--}337\times$	$13\times$ gate	$\rho=.65$

3.1 SETUP

We follow the training and measurement protocol of the companion measurement study [Zhu, 2026] exactly. Base model Qwen2.5-1.5B [Qwen Team, 2024] (weights pinned to a single commit), six tasks (math, code, instruction, safety, summarization, translation [Cobbe et al., 2021, Taori et al., 2023, Ji et al., 2024, Narayan et al., 2018, Tiedemann, 2012]) finetuned as rank-16 LoRA [Hu et al., 2022] task vectors with response-only loss, three training seeds each. Task vectors are norm-matched to the global core-median norm before composition, and merged models $\theta_0 + \alpha\Delta_A + \beta\Delta_B$ are evaluated on a 6×6 interior grid of $(\alpha, \beta) \in (0, 1.2]^2$. Each prompt stratum contains 60 fixed prompts; all hidden-state quantities are read at the final prompt token. Our running example throughout is the code+safety merge evaluated on code prompts. Every confirmatory run executes on a single A100 with deterministic attention kernels and TF32 disabled; §3.5 explains why this is not pedantry.

3.2 LAYERWISE LEDGER: CARRIED VS. GENERATED

For a prompt x , let $h_\ell^{00}, h_\ell^{A0}, h_\ell^{0B}, h_\ell^{AB}$ denote the residual state entering block ℓ under the four weight settings. The **cumulative cross-term**

$$I_\ell = h_\ell^{AB} - h_\ell^{A0} - h_\ell^{0B} + h_\ell^{00} \quad (1)$$

is everything non-additive that has accumulated before block ℓ . To separate what block ℓ *adds* from what it *transports*, we evaluate all four block functions on one **common reference state** $\bar{h}_\ell = h_\ell^{A0} + h_\ell^{0B} - h_\ell^{00}$ (the additive reconstruction) and take the same four-corner difference of the outputs:

$$G_\ell = F_\ell^{AB}(\bar{h}_\ell) - F_\ell^{A0}(\bar{h}_\ell) - F_\ell^{0B}(\bar{h}_\ell) + F_\ell^{00}(\bar{h}_\ell). \quad (2)$$

Intuitively, G_ℓ is the cross-term block ℓ would create from scratch on interference-free input; the exact ledger $I_{\ell+1} = G_\ell + P_\ell$ then attributes the remainder P_ℓ to transport of what was already carried (we do not interpret P_ℓ beyond this accounting identity). Two structural identities hold exactly and serve as built-in sanity checks: $I_1 = 0$ (LoRA leaves the embedding untouched) and $I_2 = G_1$; both are verified to machine zero in every run.

3.3 OUTPUT-SIDE ESTIMANDS

Following Zhu [2026], expressed interference on a prompt batch is the interaction ratio R : the mean Jensen–Shannon divergence between the merged model’s next-token distribution and its no-interaction additive prediction (logit-space $\ell^{A0} + \ell^{0B} - \ell^{00}$), normalized by the mean main-effect JSD—a ratio of means, never a mean of ratios. The causal effect of an intervention is

$$D = R_{\text{original}} - R_{\text{patched}}, \quad (3)$$

with the *unpatched* main-effect denominator frozen in both terms, so D moves for exactly one reason. Preregistered contrasts are paired differences of these summaries across conditions sharing prompts, evaluated with prompt-paired bootstrap CIs and 3/3 seed-direction agreement, seeds being the outer replication unit.

3.4 CROSS-TERM ERASURE AND ITS CONTROLS

The intervention replaces the merged model’s residual state at a frozen boundary with $h_\ell^{AB} - \lambda I_\ell$: at $\lambda = 1$, the carried cross-term is exactly erased and the state is restored to the additive reconstruction; the ladder $\lambda \in \{0, 0.5, 1, 1.5, 2\}$

probes dose dependence. The claim “the *direction* of I_ℓ is what matters” is only as strong as its controls, so every control preserves the displacement norm $\|I_\ell\|$ per prompt while destroying exactly one structural property, and we report each control’s overlap $\cos(\delta, -I_\ell)$ so that direction-mediation can be checked rather than assumed:

- **wrong-pair**: the additive state borrowing the second axis from the *other* task pair (kills: “any task-structured direction works”);
- **coefficient-mismatch**: the $(+1, -1, +1)$ combination in the same affine span (kills: “any span-consistent displacement works”);
- **cross-prompt**: the additive state of a *donor* prompt, assigned by a frozen length-binned derangement, applied as a full dose ladder (kills: “prompt identity is irrelevant”; the direct null for input-conditioning);
- **distance-matched random** (kills: “perturbation per se helps”).

Interventions run in float32; damage is audited at two levels—first-token (top-1 agreement, entropy shift, JSD from unpatched) and prefill-patched greedy generation on both endpoint task strata (token agreement; reference-answer containment for math).

3.5 NUMERICAL VALIDITY, PREREGISTRATION, AND WHY BOTH WERE LOAD-BEARING

The four-corner difference is a catastrophic cancellation—the four paths differ at $O(\alpha)$ while I_ℓ is $O(\alpha\beta)$ —so every estimator in this paper carries its own noise floor and gates: (i) a *determinism null* (four forwards at identical coefficients must agree bitwise; enforced, not assumed); (ii) a *propagated-ULP floor* (half-ULP representation jitter on the four stored states, propagated in float64 through the cancellation; any layer below its floor is marked uninterpretable); (iii) finite-difference probes carry step-halving convergence checks; (iv) the exact identities of §3.2. These gates are not decoration: they caught, before any conclusion depended on them, an all-bf16 mixed-derivative estimate that was pure noise, a floor computation silently zeroed by float32 rounding of sub-ULP jitter, and ultimately the §4.1 finding that naive bf16 G is mostly weight-rounding roughness.

All hypotheses, pass/fail criteria, layer positions, dose ladders, control constructions, and the three §5–6 predictions were frozen—in a versioned freeze log with a full change record—before the corresponding data existed; falsified predictions (two of three headline expectations among fourteen frozen predictions overall) are reported as falsified, with their prior confidences. Verdicts, immutable per-cell results, and the full freeze chain are released (Reproducibility Statement).

4 TRANSPORT DOMINATES GENERATION—AND MAGNITUDE IS NOT THE SIGNAL

We begin with the question the field’s metrics implicitly answer: *how much* cross-term does each layer produce, and does that quantity carry the functional differences?

Measured naively at training precision, local generation looks like a universal constant—a result that dissolves under audit and is instructive precisely because of how it dissolves. Across all twelve (pair $\times$ stratum) conditions of our main sweep—1,296 preregistered cells spanning two task pairs, six prompt strata, three seeds, and the full composition grid—the bf16 full-depth generation summary $AUC(G)$ occupies a band of 0.0101–0.0116 (per-pair ratios 1.02–1.15 $\times$), and the band even holds for the geometrically most conflicting and most orthogonal pairs selectable from the adapter gram matrix. The frozen precision audit (§4.1) attributes this uniformity to a task-independent quantization floor. Recomputed with float32 blocks over the full grid (36 cells, preregistered), the real picture is different: the fp32 band spans 3.4 $\times$ (0.0012–0.0040, falsifying our own frozen $<2\times$ expectation), the pair ratio is 1.81–1.90 $\times$ —*coarsely tracking* the $2\times$ output difference, as every magnitude quantity in this paper turns out to (§7)—and generation under instruction wrappers is 1.7–1.8 $\times$ larger than raw, 3/3 seeds.

The output side sets the standard these numbers must meet. The two pairs differ by roughly $2\times$ in expressed interference (family scores 10.33% vs. 5.31%) and—causally, by erasure—14 $\times$ –337 $\times$ in removable interference (§5); the same prompts differ by roughly 20 $\times$ across formats (+6.91pp raw vs. +0.34pp wrapped) [Zhu, 2026], in the *opposite* direction from generation. **Generation magnitude is at best a coarse correlate of pair structure and is inverted on the format axis—we call the assumption that it carries the functional differences the *magnitude fallacy*** (Figure 1a).

4.1 A CAUTIONARY RESULT: NAIVE BF16 ESTIMATES OF G ARE MOSTLY ROUND-OFF

Before interpreting the uniform band, we report what it is made of. Recomputing G_ℓ on the *same* common reference state with the block cast to float32 shrinks it to 14–23% of its bfloat16 value, with directions nearly uncorrelated between precisions (median cosine 0.15–0.24). **75–90% of the naively measured “local generation” is bfloat16 weight-rounding roughness, not smooth-function structure**—and rounding roughness is task-independent, which is precisely *why* the band is universal. Two properties of our pipeline surfaced this: every estimator carries its own propagated-ULP noise floor, and every finite-difference probe carries a convergence check (§3.5). We flag this as a standalone caution: activation-difference analyses run at training precision can be dominated by quantization roughness while passing casual sanity checks. Paired *contrasts* survive—the rounding floor is condition-independent and cancels in differences—and all causal results in §5 are computed in float32.

4.2 TRANSPORT DOMINATES: AN EXACT DECOMPOSITION, PLUS THE INJECTION BOUNDS

The accounting identity $I_{\ell+1} = G_\ell + P_\ell$ leaves P_ℓ uninterpreted, and rightly so: it mixes transport of the existing term with state–parameter coupling. We therefore decompose it *exactly*. Evaluating all four blocks (float32) both at the common reference $\bar{h}_\ell$ and at the four actual path states regroups the actual-state four-corner term, bitwise, as

$$I_{\ell+1} = G_\ell + \underbrace{F_\ell^{AB}(h_\ell^{AB}) - F_\ell^{AB}(\bar{h}_\ell)}_{T_\ell \text{ (transport of the existing } I_\ell)} + M_\ell, \quad (4)$$

where M_ℓ bundles the marginal-path state mismatches (the state–parameter coupling channel); $G_\ell + M_\ell$ is precisely the one-step regeneration a block produces when I_ℓ has just been erased. Over 36 preregistered cells: **transport carries ~69% of the layerwise flux** (regeneration share 0.31, 3/3 seeds), late-network transport exceeds regeneration by 2.3 $\times$, and **transport alone amplifies**— $\|T_\ell\|/\|I_\ell\| = 1.08$ per late block, 3/3 seeds (both predictions frozen in advance; Appendix L). A preregistered replication on Llama-3.2-1B reproduces both: transport share 65%, late gain 1.14, 3/3 seeds—transport dominance is a family invariant. The coarser arithmetic exclusion agrees: if layers contributed independent increments of the measured $\|G_\ell\|$, cumulative growth is bounded by 0.088 (incoherent) and 0.369 (perfectly aligned), against an observed $\|I_{28}\| = 0.974$ —**2.6 $\times$ even the coherent bound**. Consistently, smooth local generation is small and flat—mixed-derivative stencils at five centers of the composition rectangle are level to within 1.4 $\times$ —in agreement with CTL’s approximate-linearity observation [Zhou et al., 2024], while relocating everything that observation leaves unexplained into propagation.

Refuted conjecture, reported per our preregistration: we predicted (55% prior confidence) that mixed-derivative mass concentrates off-diagonal (“one adapter at full strength, the other entering”); measured edge-to-diagonal ratios of 1.005–1.010 (max 1.37, zero blocks above the preregistered 2 $\times$ threshold) falsify it.

5 ORIENTATION IS LOAD-BEARING: CROSS-TERM ERASURE

Section 4 shows magnitude cannot carry the differences. This section shows what does—causally. We intervene at five frozen depths with cross-term erasure and the four norm-matched structural controls of §3.4, with all three predictions registered before implementation.

Erasing the cross-term direction removes interference; every matched control fails—and the structured ones backfire. At $\lambda = 1$ on code prompts (code+safety, mean over five depths and three grid points), $D_{\text{correct}} = +0.0301/+0.0289/+0.0160$ across seeds—removing roughly 19% of expressed interference (Table 2; a depth-average that the regrowth analysis below decomposes: early erasures contribute almost nothing, late erasures nearly everything). The preregistered specificity contrasts are positive 3/3 seeds against *every* control: +0.049 to +0.059 vs. wrong-pair, +0.058 to +0.072 vs. coefficient-mismatch, +0.026 to +0.046 vs. cross-prompt, +0.021 to +0.037 vs. random. The wrong-pair and coefficient-mismatch patches in fact have *negative* D : an equally large, equally structured displacement in the wrong direction makes interference worse. Conditioning on each control’s overlap $\cos(\delta, -I_\ell)$, the effect is proportional to the displacement’s projection on the erasure direction (e.g., at depth 14: correct $\cos=1.00$, $D=+0.028$; wrong-pair $\cos=0.25$, $D=+0.007$), so we claim *cross-term-direction specificity*—the direction is the mediator-facing variable—rather than task-identity specificity beyond it.

Dose–response dissociates direction from displacement. Along the correct direction, $D(\lambda)$ rises and saturates: $0 \rightarrow +0.017 \rightarrow +0.025 \rightarrow +0.026 \rightarrow +0.020$ for $\lambda = 0, 0.5, 1, 1.5, 2$. Along the norm-matched cross-prompt direction it *worsens monotonically*: $0 \rightarrow -0.001 \rightarrow -0.012 \rightarrow -0.034 \rightarrow -0.066$ (Figure 1b). No account in which perturbation per se helps or hurts survives both curves: the sign of the effect is decided by direction alone, and the correct direction has a natural completion point near $\lambda \approx 1$, exactly where the cross-term is fully erased.

Causally removable interference differs by more than an order of magnitude between pairs whose generation magnitudes differ by less than $2\times$. Returning to the running example: on the very code prompts where the two merges’ local generation differs by at most $1.9\times$ in float32 (§4), erasure removes substantial interference from code+safety and almost nothing from code+math— $D = +0.0022/+0.0001/+0.0005$ across seeds, giving pair ratios of $13.9/337/29.5$ (the middle value is inflated by a near-zero denominator; the direction is unambiguous; in *unnormalized* JSD the absolute reductions differ by $18\text{--}52\times$ with one seed’s code+math reduction indistinguishable from zero). This is the paper’s central quantitative contrast: **the pair difference lives overwhelmingly in the orientation of the carried cross-term, far beyond what its size can explain.**

Removal is not destruction. Erasure at $\lambda = 1$ leaves first-token behavior almost untouched: top-1 agreement $0.94\text{--}0.99$ across pairs and seeds, small entropy shifts, patched-state JSD from unpatched (0.003) two orders below the main effect (0.236); prefill-patched generation preserves $92\text{--}99\%$ of continuation tokens and math containment stays at 0.83.

Erasure is undone by propagation—except where depth runs out: the carried term is an attractor. If interference were *stored* in the cross-term, erasing it should remove it for good. We preregistered a follow-up measurement (window and priors frozen before data; Appendix G) tracking the cross-term downstream of a $\lambda=1$ erasure, expecting partial regrowth ($\rho = \|I^{\text{post}}\|/\|I^{\text{orig}}\| \in (0.3, 0.95)$ at the output). The window was *falsified from above*: erase at depth 3 or 8 and the network rebuilds the cross-term essentially exactly—final-boundary $\rho = 1.00/0.99$, direction cosine to the original cross-term 0.99 (the frozen direction-memory prediction, $\cos > 0.5$, held with room to spare). Regrowth only falls off when depth runs out: $\rho = 0.94$ erasing at 14, 0.84 at 20. And this profile interlocks with the causal effect: decomposing D by patch depth, early erasures achieve nothing ($D \approx 0$ at depths 3 and 8, 3/3 seeds) while late erasures carry the entire effect ($+0.026$ to $+0.058$ at depths 20–25)—removable interference is exactly what the remaining depth cannot rebuild. The reading we draw (formed after these results; its own predictions are untested): the carried cross-term is an *attractor* of the forward pass, continuously re-established from the untouched marginal paths rather than stored and read out. This sharpens §4.2’s exclusion into an interventional statement—propagation does not merely amplify the carried term, it actively reconstructs it in norm and direction—and yields a design corollary: erasure-style mitigation must act late, where reconstruction cannot catch up.

The depth profile rules out an estimand artifact. Intervention target and output estimand share the additive counterfactual, so the reduction might be partly *by construction*. The depth decomposition refutes this: a construction-driven effect would appear at every depth—restoration is as thorough at depth 3 as at 25—yet $D \approx 0$ at depths 3–8, concentrating where regrowth runs out of layers, in proportion to $1 - \rho$; the coefficient-mismatch control, in the same affine span as the target, *backfires*. *The claim does not extend to behavioral quality, and we tested this.* A continuous endpoint—the merged model’s teacher-forced NLL of frozen expert continuations, independent of the additive counterfactual—passes its preregistered validity gate (merge—expert $+0.011$ to $+0.026$, 3/3) and returns a *negative* result: erasure moves it by ± 0.002 with inconsistent sign, and wrong directions are not worse (Appendix I). Erasure controls the non-additivity of the output distribution, not expert-likeness of behavior—a boundary we state as a result.

Statistical basis, disclosed: the original runs persisted per-condition scalars, so the frozen verdict rests on 3/3 seed-direction consistency with order-of-magnitude margins. A full re-run with per-prompt persistence—reproducing every immutable scalar bitwise—supplied the frozen-template bootstrap (5,000 paired draws/seed): 27 of 28 intervals exclude zero; the one containing zero, D_{wrapped} for one seed, is itself the frozen PK2-2 expectation (Appendix B). A two-level seed $\times$ prompt bootstrap agrees: $D_{\text{correct}} = +0.018 [+0.010, +0.031]$, contrast vs. wrong-pair $+0.033 [+0.021, +0.049]$.

6 EXPRESSION GATING: FORMATS HIDE INTERFERENCE WITHOUT REMOVING IT

The magnitude fallacy has a second, sharper casualty: the assumption that what an evaluation *sees* tracks what the model *carries*. We test it with the format bridge—the same 60 public code prompts presented raw and wrapped in an Alpaca instruction template—where the companion study established the output-side fact: expressed interference collapses from $+6.91\text{pp}$ to $+0.34\text{pp}$ under the wrapper, a $20\times$ contraction [Zhu, 2026].

Instruction wrappers amplify internal interference while collapsing its expression. We preregistered the natural expectation—internal generation should shrink with the output (60% prior)—and the data reversed it: wrapping *increases* local generation, $C_{\text{format}} = -0.00075$ [CI $-0.00080, -0.00070$] (code+safety) and -0.00061 [$-0.00067, -0.00055$] (code+math), 3/3 seeds, companion agreeing in sign. The template is in-distribution for both adapters, so both axes

Table 2: **Cross-term erasure is direction-specific: the correct direction removes interference, every norm-matched control fails or backfires** ($\lambda=1$, code prompts, code+safety, mean over 5 frozen depths $\times$ 3 grid points; per-seed values). Brackets: 95% prompt-bootstrap CIs (5,000 paired draws per seed, frozen template) from the per-prompt re-run, which reproduced every immutable scalar bitwise; all 15 intervals exclude zero.

	seed 42	seed 123	seed 456
$D(\text{correct})$	+0.0301 [.021, .042]	+0.0289 [.023, .037]	+0.0160 [.013, .020]
$D(\text{correct})-D(\text{wrong-pair})$	+0.0556 [.042, .074]	+0.0587 [.044, .077]	+0.0487 [.040, .059]
$D(\text{correct})-D(\text{coeff-mismatch})$	+0.0699 [.053, .091]	+0.0719 [.056, .091]	+0.0583 [.048, .069]
$D(\text{correct})-D(\text{cross-prompt})$	+0.0460 [.038, .055]	+0.0391 [.032, .047]	+0.0257 [.021, .031]
$D(\text{correct})-D(\text{random})$	+0.0375 [.027, .050]	+0.0355 [.029, .044]	+0.0206 [.017, .025]
$D(\text{correct}), \text{code+math}$	+0.0022	+0.0001	+0.0005

activate and their mixture grows—while the template pins the output distribution and the grown interference never surfaces. We call this dissociation *expression gating*, and report the reversal as a falsified prediction whose opposite replicated 3/3.

Erasure confirms the gate causally: where interference is not expressed, there is nothing to remove. If the wrapper merely *masked* measurement noise, erasure should work equally under both formats. It does not. The identical $\lambda=1$ erasure yields $D_{\text{raw}} = +0.0286/+0.0370/+0.0174$ across seeds but $D_{\text{wrapped}} = +0.0031/+0.0026/+0.0005$ —a $13\times$ mean ratio, raw>wrapped 3/3 (Figure 1c). The wrapped condition has almost no removable interference because almost none is expressed: the wrapped merged model’s interference ratio is itself only $R \approx 0.003$. (At 7B the same measurement finds the gate fully closed: D_{wrapped} indistinguishable from zero, all seeds, Appendix K.)

What the gate is made of: the wrapper drowns the interaction rather than shrinking it. Decomposing the frozen ratio, the collapse is denominator-driven: the wrapper inflates the main-effect JSD $\sim 13\times$ ($0.025 \rightarrow 0.32$) while the *absolute* interaction JSD does not shrink (raw/wrapped $0.4\text{--}1.0\times$; it is, if anything, larger wrapped—the output-side counterpart of the internal amplification above), and the absolute JSD removed by erasure is format-comparable. The gate is therefore not destruction of the interaction but *relative and behavioral drowning*: the template pins the output distribution, and an interaction of similar absolute size becomes negligible against it—which is exactly what the normalization-free behavioral collapse measures ($+6.91\text{pp} \rightarrow +0.34\text{pp}$ [Zhu, 2026]). The structure generalizes across two further instruction templates (denominator $11\text{--}35\times$, R down $18/18$) and is absent for a length-matched non-instruction prefix, which inflates numerator and denominator together (Appendix J). We state the $13\times/20\times$ figures in the frozen relative estimand throughout, with this decomposition as their reading.

Exploratorily, the gate is a depth-relocation of amplification. The wrapper front-loads amplification (early gain 1.55 vs. 1.36 raw; raw–wrapped late-gain contrast positive 3/3) and shows the largest super-coherent amplification ($3.8\text{--}4.1\times$ vs. $1.0\text{--}1.7\times$ raw): template tokens amplify the carried term early, and the pinned distribution absorbs it before expression (Appendix E; exploratory, computed after the confirmatory verdicts).

Implication for evaluation practice. A merged model can carry *more* internal interference precisely when a format-wrapped behavioral evaluation reports it *cleanest*. Within the scope we have measured—one wrapper family, two core task pairs, with the direction (though not the strength) of the effect replicating at all three model points—an assessment of merged models that only reads wrapped outputs is reading the gate, not the interference. This is the mechanism-level counterpart of behavior-level format fragility reported by RAIN-Merging [Huang et al., 2026]. We deliberately make no claims about safety *behaviors* here; the corrected belief is about what evaluations measure.

7 WHAT EXISTING METRICS SEE—AND DON’T

Our final section closes the loop with the metrics the field already uses, evaluated on their own terms: can they *predict* interference for unseen task pairs, and can they *explain* the causal structure of §5–6? For all 15 pairs of six tasks (6 calibration pairs among the original tasks, 9 held-out pairs involving two unseen tasks), we compute CTL linearity and relative residual, parameter-space cosine overlap, our preregistered direction features, and cumulative cross-term magnitude—against measured output interference, under a frozen protocol in which only feature *signs* are fitted on calibration pairs.

Linearity per se is uniformly high and separates nothing. CTL’s central observation replicates *more strongly than reported*: layerwise linearity cosines are $0.93\text{--}0.98$ across every family, and in float32 the local nonlinear generation is smaller still (§4.1). But precisely because it is uniform, the linearity level cannot explain why one pair interferes at $2\times$ another, or why a wrapper changes expressed interference $20\times$. Parameter cosine fares little better as a predictor

(held-out Spearman $\rho = +0.25$, 4/8 on the preregistered high-vs-low bins).

SurgeryV2’s representation bias behaves like cumulative magnitude: coarse on pairs, blind to the gate. Measured against the marginal experts on all 36 cells (two predictions frozen first, Appendix H), it orders the pair contrast ($2.9\times$, 3/3—expected, since bias contains the amplified cumulative term), but on the format bridge, where expressed interference collapses $\sim 95\%$, bias moves between -1% and $+34\%$ (frozen “wrapped $\geq$ raw” held 5/6, one sub-percent inversion reported as required). Every magnitude-family quantity we measure sees at most coarse pair structure; none sees the gate.

Cumulative magnitude coarsely predicts pair ordering—because amplification differs across pairs. Cumulative $\|I\|$ attains held-out $\rho = +0.63$ – 0.65 , 3/3 seeds, 8/8 preregistered comparisons (exact permutation $p=0.25$ over 12 assignments; prospective directional evidence, not significance)—consistent with §4: propagation gain is pair-dependent, so *post-amplification* magnitude carries coarse pair information. It remains blind to the two sharper facts: the causal pair contrast exceeds what cumulative magnitude spreads can explain (§5), and on formats every magnitude points the *wrong way*—the wrapped condition carries more internal cross-term yet exposes $13\times$ less removable interference (§6). Magnitude, cumulative or local, cannot see the gate.

Our own preregistered direction features failed, and we report the autopsy. None of three frozen features met its pass criterion; the best ($\rho = +0.62$) was contaminated—its logit-lens normalization makes it $\propto 1/\|I\|$, a magnitude signal in disguise—and the others showed no held-out signal. Cheap directional predictors remain open; direction’s causal role (§5) is established by intervention, not by these features.

8 DISCUSSION, LIMITATIONS, AND CONCLUSION

What the intervention establishes. Effects are responses to norm-matched interventions, proportional to the projection on the cross-term direction: we claim direction-mediation under intervention, not natural mediation [Vaidyanathan et al., 2026]; $\lambda > 1$ states are off-manifold and only the preregistered ordered contrasts are interpreted there.

Second-family replication. Exercising the freeze chain, we replicated the reduced protocol on Llama-3.2-1B (different pretraining data, tokenizer, 16 blocks vs. 28, GQA), with four predictions frozen before any adapter on that family existed (Appendix F). All four held: the G band ratio is 1.20 (gate < 1.3) while output interference again varies beyond it; the injection exclusion is *stronger* (cumulative cross-term 2.2 – $7.9\times$ the coherent bound); erasure is direction-specific with all 15 bootstrap intervals excluding zero, and removable interference again separates pairs 2.6 – $5.5\times$ where generation magnitudes differ by $\leq 1.2\times$. Two honest qualifications: the output-side pair and format gaps are directionally consistent (3/3 seeds, including the $G_{\text{wrapped}} > G_{\text{raw}}$ reversal) but smaller and seed-variable, and expression gating is correspondingly weaker (raw/wrapped removable interference $\approx 2\times$ vs. Qwen’s $13\times$, one seed’s contrast CI crossing zero). The mechanism transfers; the *magnitudes*—especially the strength of the gate—are family properties, and the $13\times/20\times$ figures are scoped to Qwen2.5-1.5B throughout. A third grid point at larger scale—Qwen2.5-7B, reduced protocol, predictions again frozen first (Appendix K)—completes the picture: direction-specificity and pair-carrying replicate with all bootstrap intervals excluding zero (pair ratios 6.9 – $22.9\times$), the injection exclusion strengthens (up to $8.3\times$ the coherent bound), and gating is *total* at 7B (D_{wrapped} indistinguishable from zero, 3/3), while the G -band gate missed its frozen threshold by 0.1% —reported as the letter requires. Across the three model points the mechanism is invariant and every *magnitude* is not: output mismatch spans 1.5 – $8.6\times$, gating spans $\sim 2\times$ to fully closed.

Limitations, each with its direction. (i) *Three model points, one adapter parameterization* (Qwen2.5-1.5B full; 7B and Llama-3.2-1B reduced; rank-16 LoRA). Full finetuning and further families remain open; the magnitude variation across points is unexplained; we avoid “universal.” (ii) *Intervention statistics initially rested on 3/3 seed-direction agreement alone*; the per-prompt re-run added bitwise replication and bootstrap CIs (27/28 excluding zero, §5), but CIs are within-seed and three seeds remain the outer unit. (iii) *Controlled prompts are diagnostic instruments*; gating evidence rests on one wrapper template and two core pairs, the primary estimand is first-token, and we make no safety-behavior claims. Behavior-level validation was attempted three times and reported in full (Appendix I): discrete scoring was a null instrument at both scales, and a continuous NLL endpoint passed its validity gate but *falsified* behavioral recovery—the paper claims nothing beyond expressed non-additivity. (iv) *Our preregistered direction features failed on held-out pairs* (§7); cheap directional predictors remain open, and erasure as a mitigation method is unevaluated.

Conclusion. Forty-six predictions were frozen before their data—fourteen in the main study, thirty-two across replications and follow-ups, every miss reported by the letter of its frozen criterion—and the mechanism that survived is not the one we set out to confirm. Interference in weight-space composition is not a quantity to be measured but a direction to be traced: transported and amplified, restored when removed, expressed only where the format allows.

Reproducibility statement

All hypotheses, thresholds, layer positions, dose ladders, and control constructions were frozen before the corresponding data existed, in a versioned preregistration (six freeze versions, each superseding change logged); all fourteen predictions and their outcomes, three verdict documents, 1,296+90+54 immutable per-cell results plus the 54 per-prompt intervention records and their bootstrap-CI report, the second-family replication preregistration, verdict, and its 54+54 cells, the frozen regrowth follow-up and its 18 cells, the frozen SurgeryV2-bias comparison and its 36 cells, the frozen scale point and its 54+54 cells, the frozen transport-decomposition/fp32-grid experiment and its 36 cells, the numerical-validity audits (determinism nulls, ULP floors, bitwise equivalence audits), the exact software stack, and the pinned base-model commit are released with the code. Every figure is generated by a script that recomputes its numbers from the immutable cell records.

Statement on AI usage

Large language models were used, under continuous author direction and review, to assist with experiment-code implementation, data-analysis scripting, and manuscript drafting and editing. All hypotheses, preregistered predictions, pass/fail criteria, and verdicts were specified or approved by the authors; every reported number derives from immutable experiment records released with the code, and all figures are regenerated from those records by released scripts. No LLM was used to generate, impute, or select experimental data.

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A FULL ESTIMATOR AND AGGREGATION DEFINITIONS

All definitions below are transcribed from the frozen design document (§B); none was altered after data existed.

Four weight-state paths. For task vectors Δ_A, Δ_B and coefficients α, β , we instantiate four weight states $M_{00}, M_{\alpha 0}, M_{0\beta}$, and $M_{\alpha\beta}$ and, for a prompt x , record the residual state at the input of block ℓ under each: $h_\ell^{00}, h_\ell^{\alpha 0}, h_\ell^{0\beta}$, and $h_\ell^{\alpha\beta}$. All hidden analyses use the *last prompt token* only, aligning with the companion study’s first-generated-token logit proxy and avoiding position-averaging bias between raw and wrapped sequences of different lengths.

Cumulative cross-term (descriptive only). $I_\ell = h_\ell^{\alpha\beta} - h_\ell^{\alpha 0} - h_\ell^{0\beta} + h_\ell^{00}$. Because I_ℓ contains every nonlinearity produced and propagated by blocks $1..\ell-1$, depth curves of I_ℓ are frozen as descriptive: they cannot by themselves locate where cross-terms are *generated*.

Common-state local generation (KT1 primary object). With F_ℓ^{ab} the full block- ℓ map (residual update included) of weight state M_{ab} , and the additive reference $\bar{h}_\ell = h_\ell^{\alpha 0} + h_\ell^{0\beta} - h_\ell^{00}$,

$$G_\ell(\bar{h}_\ell) = F_\ell^{\alpha\beta}(\bar{h}_\ell) - F_\ell^{\alpha 0}(\bar{h}_\ell) - F_\ell^{0\beta}(\bar{h}_\ell) + F_\ell^{00}(\bar{h}_\ell).$$

Because all four block paths receive the identical input, G_ℓ isolates the four-corner term contributed by the combined weights at the current reference state. The exact ledger $I_{\ell+1} = G_\ell + P_\ell$ defines the *state-conditioned remainder* P_ℓ , which is frozen as *not* interpretable as pure propagation (it mixes propagation of existing interaction, reference-state mismatch, state-parameter coupling, and higher-order block nonlinearity).

Profiles, AUCs, and companions. The primary layer profile is $g_\ell^{\text{RMS}} = \text{RMS}(G_\ell)$ with full-depth AUC as the headline summary (early-depth AUC over normalized depth $[0, 0.5]$ is a frozen secondary). The functionally weighted companion applies $\phi(z) = W_U \text{FinalNorm}(z)$ to each corner before differencing; untuned logit-lens is confirmatory only at normalized depth $d \geq 0.5$, descriptive earlier. The two frozen contrasts are $C_{\text{pair}} = \text{AUC}(G; \text{code+safety}) - \text{AUC}(G; \text{code+math})$ and $C_{\text{format}} = \text{AUC}(G; \text{raw}) - \text{AUC}(G; \text{wrapped})$.

Output estimand and intervention effect. R is the companion study’s ratio-of-means (never mean-of-ratios):

$$R = \frac{\text{mean}_x \text{JSD}(p_x^{\alpha\beta}, p_x^{\text{add}})}{\text{mean}_x \text{JSD}(p_x^{\alpha\beta}, p_x^{00})}, \quad D_\ell^{(q)} = R_{\text{original}} - R_{\text{patched}, \ell}^{(q)},$$

with the denominator of R_{patched} frozen to the *unpatched main effect*—a preregistered choice: the patched model’s own main effect also moves, and a self-normalized ratio would confound two movements in one number. Unnormalized-JSD shrinkage is a clearly-separated descriptive secondary. R/D are formed once from the concatenated prompt set (micro-batch averaging of D is forbidden by the freeze).

Aggregation order and statistics. Track A computes every hidden estimand on the same norm-matched 7×7 composition grid’s 36 interior points, prompts, seeds, and aggregation order as the companion study; KT2 uses the frozen subset $\{(1, 1), (0.6, 1), (1, 0.6)\}$ and five frozen boundaries $\{3, 8, 14, 20, 25\}$. Seeds are the preregistered outer unit (3/3 direction agreement required); within seed, paired prompt bootstrap with 5,000 draws (frozen RNG seed 20260712) yields the CIs reported for KT1 contrasts.

B PREREGISTRATION TIMELINE AND THE FOURTEEN PREDICTIONS

Freeze chain. The design document was versioned v1.0–v1.6; every version’s full text is retained verbatim, and any change to hypotheses, gates, units, doses, statistics, or the decision tree produced a new version rather than a silent overwrite. Table 3 lists the chain. Versions v1.0–v1.4 all predate any hidden-state result. v1.5 is explicitly a *post-KT1* revision—it does not rewrite the KT1 verdict (archived under v1.4 rules) but freezes the hypotheses and pass criteria for the *next* experiments (KT2, direction features, CTL comparison, adversarial arm) before any of their data existed, and adds the “stable reversal” rule the original decision tree lacked. v1.6 adds the bf16 dtype audit with its reading rules frozen before the audit ran. A separate point-prediction memo (P1–P6) was frozen after two pilot cells but before the 1,296-cell sweep launched.

Version	Content of the change
v1.0	Initial freeze (2026-08-02): hypotheses, estimators, units, KT1–KT3 gates, decision tree.
v1.1	D denominator fixed to unpatched main effect; determinism null split from ULP floor; patch-quantization audit added. Pre-data.
v1.2	Hardware lock (single A100, sdpa, deterministic, TF32 off); precision lock (bf16 KT1/KT3, fp32 KT2). Pre-data.
v1.3	Batching lock (micro-batch 8, global pad width); aggregation order; equivalence-audit requirement. Pre-data.
v1.4	Per-stage execution manifests (restores the frozen staged order; relaxes no gate). Pre-data.
v1.5	Post-KT1: PK2-1/2/3 frozen before KT2 implementation; F1–F3 and CTL-table analysis preregistration; adversarial arm; stable-reversal rule.
v1.6	bf16 G dtype audit with frozen reading rules and prior (55% “ G real”).
P1–P6 memo	Point predictions with priors, frozen before sweep launch.

Table 3: The preregistration freeze chain. Full documents and sidecars are released with the code.

The fourteen predictions. Table 4 lists every frozen prediction with its prior confidence and outcome: 7 held, 4 failed, 3 split. The additionally frozen dtype-audit prior (55% that bf16 G was real signal) also fell on the wrong side (§4.1); it is listed separately as an audit belief rather than an experimental prediction.

#	Prediction (frozen)	Statement	Prior	Outcome
1	P1-direction	$C_{\text{pair}} > 0$, both strata, 3/3 seeds	70%	held
2	P1-magnitude	AUC ratio in 1.3–2.0	70%	failed (1.02–1.15)
3	P2	math-stratum contrast $< \frac{1}{2}$ of code, still positive	high	held
4	P3	raw $G > \text{wrapped } G$, ratio > 1.5	60%	failed (<i>reversed</i> , 3/3)
5	P4	generation share 4–8%	high	split (carried-dominance holds; window wrong: mean 12.6%)
6	P5	$\cos(G, B)$ depth profile	high	split (shape yes; variance far larger)
7	P6	all numerical gates green	high	held (1,296/1,296)
8	PK2-1	erasure direction-specific vs. all controls	65%	held (3/3)
9	PK2-2	expression gating: $D_{\text{raw}} \gg D_{\text{wrapped}}$	55%	held (3/3, 13×)
10	PK2-3	pair contrast carried by direction, ratio > 1.3	50%	held (14×–337×)
11	CTL separation	no CTL/TSV metric separates pair or format ordering	70%	split (cumulative $\ I\ $ <i>does</i> order pairs; format still invisible)
12	F1–F3	≥ 1 direction feature passes held-out gate	45%	failed (0/3; F1 magnitude-contaminated)
13	Adversarial band	extreme-geometry pairs stay in G band	60%	held
14	Off-diagonal mass	edge stencil $ B > 2 \times \text{diagonal}$	55%	failed (max 1.37)

Table 4: All frozen predictions and outcomes. “Split”: qualitative claim held, quantitative window failed. No outcome was rewritten after data; reversals are reported as falsifications with the reversed reading tested by *new* frozen predictions (rows 8–10).

C NUMERICAL VALIDITY AUDITS

The four-corner difference is a catastrophic cancellation: the four paths differ at $O(\alpha)$ while I_ℓ is $O(\alpha\beta)$. Every estimator therefore carries its own validity chain, all items frozen before confirmatory execution.

Determinism null (must be exactly zero). Pointing all four weight states at the same model and running four independent forwards must produce a *bitwise-zero* four-corner difference under deterministic kernels—this validates determinism only and is explicitly *not* a noise floor (under a deterministic backend it is identically zero and can prove nothing about resolution). All 1,296 confirmatory cells passed with exact zeros.

Propagated-ULP cancellation floor (the real noise floor). The resolution limit is set by storage precision: we perturb each of the four stored hidden states by $\pm \frac{1}{2}$ ULP representation uncertainty and re-form the four-corner difference, reporting per-layer floor RMS (mean/p99/max) under frozen draws and RNG seed. One implementation subtlety is instructive: propagating the jitter in float32 *silently returns zero*—sub-ULP perturbations round back to the original value 100% of the time—so the propagation is done in float64 and validated by the exact 2^{15} ratio between fp32 (23 mantissa bits) and bf16 (8 mantissa bits) floors. At the main composition point (1, 1) the signal-to-floor ratio is min 10.2 / median 38.2 / max 92.7 across confirmatory layers; layers below ratio 1 would be flagged *below numerical resolution* and barred from mechanistic interpretation (none were). Boundary 0 (embedding) is excluded as a definitional zero, not a resolution failure.

First-block identities. With LoRA leaving the embedding untouched, $I_1 = 0$ exactly and $I_2 = G_1$ to within the ULP floor. Both identities are asserted per cell; they jointly validate the hook boundary, the reference-state construction, and the block-map definition.

Bitwise equivalence audit of performance refactors. Single-GPU feasibility required three optimizations—capture-and-replay direct block invocation, prompt micro-batching, and (initially) `logits_to_keep=1`. Each had to prove semantic invariance *on the locked A100* (kernel behavior is part of the estimand; equivalence does not transfer across hardware). The audit caught two real issues before any confirmatory data existed. (i) `logits_to_keep=1` changes the lm-head GEMM shape and hence accumulation order, producing a 1-ULP deviation (max 0.0625 at $|\text{logit}| \in [16, 32)$) against the full-sequence path; the optimization was removed rather than tolerated. (ii) `output_hidden_states` returns the *final-norm-transformed* state at the last boundary, not the last block’s raw output—mixing spaces in the ledger (observed direct-block discrepancy 67.94). Fixed with a forward hook capturing the raw output, a uniform

`residual_state()` accessor, and a regression test locking the contract. After both fixes, all equivalence checks are exactly 0.0 at every audited depth.

The bf16 dtype audit (frozen reading rules). Recomputing G_ℓ on the identical $\bar{h}_\ell$ with block weights and arithmetic cast to float32: fp32/bf16 RMS ratio median 0.230 (code+safety) and 0.136 (code+math), minima 0.073/0.040; direction cosines median 0.239/0.152, minimum -0.031 . Both frozen reading rules place this in the “ G is predominantly weight-rounding artifact” branch (§4.1). A related earlier catch: an all-bf16 mixed-derivative stencil—which extracts an $O(\epsilon^2)$ signal from $O(\epsilon)$ differences, a worse cancellation than G itself—produced pure noise ($\approx 50\times$ the true value), exposed by its own frozen gates (probe convergence ≈ 1.0 everywhere; $|B| > 50|G|$ is physically impossible). All stencils were moved to fp32 block evaluation with their own floors. The general lesson we would offer practitioners: *every new activation-difference estimator needs its own propagated noise floor and convergence check, or it will discover noise.*

D CONTROL CONSTRUCTIONS AND THE CROSS-PROMPT DERANGEMENT

At each frozen boundary $\ell \in \{3, 8, 14, 20, 25\}$ the intervention replaces the merged model’s residual state h_ℓ^{AB} with $h_\ell^{AB} + d_\ell^{(q)}$ and continues the forward pass in float32. The correct erasure has displacement $d^{(\text{correct})} = -\lambda I_\ell$ per prompt. Every control is *row-wise norm-matched*: its raw displacement is rescaled per prompt to $\|I_\ell(x)\|$, so all conditions apply identically-sized perturbations and differ only in structure.

- **Identity/no-op** ($\lambda = 0$): validates that the hook mechanism itself leaves the output bitwise unchanged.
- **Wrong-pair**: the additive reconstruction built with the *other* task pair’s second axis, $d = (h^{A0} + h^{0B'} - h^{00}) - h^{AB}$, where B' is a third adapter (for code+safety the math axis and vice versa). Preserves the additive construction; destroys task-pair identity.
- **Coefficient-mismatch**: the frozen wrong-coefficient template $(+1, -1, +1)$ in the same affine subspace, $d = (h^{A0} - h^{0B} + h^{00}) - h^{AB}$, norm-matched. Same subspace, same prompts, wrong combination.
- **Cross-prompt**: the correct additive displacement computed on a *donor* prompt x' and applied to x . Donors come from a deterministic length-binned derangement frozen before implementation (master seed 20260802): prompts are binned by token length, and within each bin a derangement guarantees no prompt donates to itself while keeping donor and recipient lengths comparable. This is the direct null for prompt-conditioned structure and the second arm of the dose ladder.
- **Distance-matched random**: per-prompt Gaussian direction scaled to $\|I_\ell\|$; frozen as a secondary sensitivity control, never a substitute for the structured nulls.

Overlap accounting. For every condition we report $\cos(d^{(q)}, -I_\ell)$ and interpret specificity contrasts conditional on it (frozen rule §8.3/§8.8 of the design). The controls span overlap cleanly: correct = 1.0 by construction; wrong-pair ≈ 0.25 (its additive scaffold shares the A -axis); coefficient-mismatch, cross-prompt, and random near zero. The observed effect is approximately proportional to the projection on $-I_\ell$ (at boundary 14: correct $\cos = 1.00$, $D = +0.028$; wrong-pair $\cos = 0.25$, $D = +0.007$; its depth-averaged D is nonetheless negative)—which is exactly why the paper claims *cross-term-direction specificity* rather than task-identity specificity beyond the direction (§8).

Patch-quantization audit. Writing patched states back to bf16 could silently cancel low-dose rungs ($\| \text{bf16}(h - \lambda I) - (h - \lambda I) \| / \|\lambda I\|$ exceeding a frozen 0.3 threshold would mark a rung *not actually applied*). The v1.2 freeze resolved this preemptively by running all of KT2 in float32 (with TF32 off, since its 10-bit mantissa would defeat the purpose); the audit is retained as residual-risk monitoring and never triggered.

E AMPLIFICATION PROFILES (EXPLORATORY)

Everything in this appendix is computed from existing immutable cell records after the confirmatory verdicts; it motivated no gate and is labeled exploratory throughout.

The exclusion arithmetic (confirmatory input to §4.2). At the (1, 1) code cell, if layers contributed independent increments of the measured sizes $\|G_\ell\|$, cumulative growth would be bounded by $\sqrt{\sum_\ell \|G_\ell\|^2} = 0.088$ (incoherent) and by $\sum_\ell \|G_\ell\| = 0.369$ even under perfect alignment of every increment. Observed: $\|I_{28}\| = 0.974$ — $11\times$ and $2.6\times$

the respective bounds, implying an average multiplicative gain of ≈ 1.19 per block on the carried term. Since the bounds are arithmetic in the measured quantities, no per-layer injection account survives regardless of the increments’ provenance (including the rounding-artifact component of bf16 G , which only strengthens the exclusion by shrinking the true injections).

Layer-resolved gain. The per-block gain $\|I_{\ell+1}\|/\|I_\ell\|$ is above 1 through most of the depth, with pair- and format-dependent profile: gains are similar across the two task pairs early and diverge in the second half of the network—consistent with §7, where *cumulative* magnitude (post-amplification) coarsely orders pairs even though local generation does not. Across the 15-pair capture grid, cumulative $\|I\|$ spans a $3\times$ range (0.084–0.256) on a generation floor that is nearly pair-independent.

Super-coherent amplification by format. Defining the super-coherence ratio as observed $\|I\|$ over the coherent accumulation bound: raw prompts $1.0\text{--}1.7\times$, instruction-wrapped prompts $3.8\text{--}4.1\times$ —amplification is *strongest* exactly where expressed interference is smallest.

Depth relocation (3/3 seeds). Splitting mean gain at normalized depth 0.5: wrapped early-gain 1.55 vs. raw 1.36, while the raw–wrapped *late*-gain contrast is $+0.019/ +0.012/ +0.031$ across seeds—positive 3/3. The wrapper front-loads amplification and suppresses it late; raw amplification persists to the output. This profile was recorded (as an exploratory expectation, not a gate) *before* KT2 ran, where it correctly anticipated the causal gating result: erasure finds interference to remove on raw prompts and almost none under the wrapper (§6).

F SECOND-FAMILY REPLICATION (LLAMA-3.2-1B)

A reduced replication with the base family as the *only* varied factor: meta-llama/Llama-3.2-1B (commit-pinned; 16 blocks, GQA, 128k tokenizer), nine LoRA adapters trained with the identical recipe (datasets, samples, epochs, LoRA shape, seeds), intervention boundaries $\{2, 5, 8, 11, 14\}$ mapped from the frozen Qwen set at equal normalized depths, and a new 9-delta core-median norm radius recorded before any kill-test cell ran. Four predictions (RP1–RP4, with priors 70/75/60/high%) were frozen before any adapter on this family existed. Execution: 54 reduced KT1 cells (KT2 subgrid $\times$ {controlled code, raw, wrapped} $\times$ 3 seeds $\times$ 2 pairs), the amplification-bound analysis, and the full 54-run KT2 with per-prompt persistence, under the same numerical-validity chain—which passed unmodified on the new architecture (RP4: bitwise determinism nulls, first-block identities, zero equivalence audits).

RP1 (band + output mismatch): held, with reduced margins. The bf16 G -AUC band ratio across all 18 (1, 1) cells is 1.201 (gate <1.3 ; absolute level ≈ 0.0013 , an order below Qwen’s—hidden width and norm scale differ, and only within-band uniformity is the cross-family claim). Output interference again exceeds what the band allows: pair ratio 1.51 (gate >1.5 ; per-seed 1.59/1.82/1.06), format ratio 2.03 (gate >2 ; per-seed 2.82/1.71/1.93). Both gates pass at the frozen mean-level reading but are seed-variable and far smaller than Qwen’s $2\times/20\times$; we grade this held-with-reduced-margins. Not lost in the reduction: $G_{\text{wrapped}} > G_{\text{raw}}$ replicates 3/3 seeds—the falsified-then-reversed KT1 prediction is not a Qwen artifact.

RP2 (injection exclusion): held, more strongly than Qwen. Cumulative-to-bound ratios per condition: 6.7–24.9 $\times$ incoherent (gate >3), 2.19–7.85 $\times$ coherent (gate >1.2). The depth-relocation profile of Appendix E also transfers: super-coherent amplification is again largest under the wrapper (5.4–7.9 $\times$ vs. 2.2–3.2 $\times$ raw), and the raw–wrapped late-gain contrast is positive 3/3 seeds ($+0.042/ +0.053/ +0.070$).

RP3 (causal triple): held; the gating leg by its frozen letter only. With the frozen bootstrap template: (a) $D_{\text{correct}} = +0.0233/ +0.0333/ +0.0192$, all 15 intervals (effect and four specificity contrasts per seed) exclude zero. (c) Implied pair ratios $D(\text{code+safety})/D(\text{code+math}) = 2.84/5.46/2.56$ (gate >1.3 , 3/3), contrast CIs excluding zero—and for the seed whose *output* pair ratio is only 1.06, erasure still separates the pairs 2.6 $\times$: the direction-carried contrast is more stable than the output ratio itself. (b) $D_{\text{raw}} - D_{\text{wrapped}} > 0$ holds 3/3 by sign, satisfying the frozen criterion, but margins are weak ($+0.0052^*/ +0.0019/ +0.0039^*$; the middle CI crosses zero) and the removable-interference ratio is only $\approx 2\times$ —consistent with this family’s weak output-side format collapse. The gate’s *direction* transfers; its *strength* is a family property.

G REGROWTH AFTER ERASURE (PREREGISTERED FOLLOW-UP)

Three predictions were frozen before any regrowth data existed (priors 60/60/65%). Setup: at each frozen boundary $b \in \{3, 8, 14, 20\}$, apply the $\lambda=1$ correct erasure and continue the forward pass; at every later frozen boundary (plus the raw final-block output) form the post-patch cross-term against the *unpatched* marginals, $I_\ell^{\text{post}} = \tilde{h}_\ell^{AB} - h_\ell^{A0} - h_\ell^{0B} + h_\ell^{00}$, and report the per-prompt recovery ratio ρ_ℓ and direction cosine to I_ℓ^{orig} . Scope: 18 cells (2 pairs $\times$ 3 strata $\times$ 3 seeds, (1, 1), float32).

Outcomes. (PE1-1, *falsified*): final-boundary ρ after erasing at depth 8 was predicted in (0.3, 0.95); observed 0.987/0.998/0.992 across seeds—regrowth is essentially complete, outside the window on the “transient” side. (PE1-2, *held*): direction memory $\cos(I_\ell^{\text{post}}, I_\ell^{\text{orig}}) > 0.5$ predicted; observed 0.987/0.987/0.986. (PE1-3, fail by letter): ρ decreasing in patch depth holds in 2/3 seeds; the third inverts between depths 3 and 8 by +0.0003, a noise-level violation we report as the frozen criterion requires. Full profile (seed means): $\rho = 1.000/0.992/0.944/0.843$ erasing at depths 3/8/14/20.

Interlock with the causal effect. Decomposing the KT2 effect by patch depth from the per-prompt records (code+safety, code stratum, mean over grid): $D \approx 0$ at depths 3–8 (+0.0005/ − 0.0012, +0.0009/ + 0.0037, +0.0001/ − 0.0076 per seed), rising to +0.0374/ + 0.0408/ + 0.0260 at depth 20 and +0.0545/ + 0.0582/ + 0.0366 at depth 25. The unrecovered share $1 - \rho$ and the durable output effect D rise together: what erasure removes is what the remaining depth cannot rebuild. The attractor reading was subsequently subjected to a preregistered basin test (gate frozen first: final-boundary $\cos > 0.8$ from every start): patching at depths 8 and 14 with correct erasure at $\lambda \in \{1, 2\}$ and all four norm-matched structural controls, the downstream cross-term reconverges to the original direction from *every* start—per-condition means 0.90–0.99, worst single cell 0.83, 18/18 (both pairs, 3/3 seeds). Convergence of direction is compatible with the divergent *output* effects of §5: wrong-direction patches commit their damage to the expression pathway before late-network regrowth completes, exactly as the depth decomposition shows.

H SURGERYV2 REPRESENTATION BIAS, HEAD-TO-HEAD

Two predictions frozen before any bias data existed (priors 70/50%). Operationalization: in the factorial setup the task experts are the marginal paths, so layerwise bias is $\|h_\ell^{AB} - h_\ell^{A0}\|$ (and against the right expert, $\|h_\ell^{AB} - h_\ell^{0B}\|$), last prompt token, bfloat16 measurement lock, full-depth AUC over the frozen boundaries; 36 cells (2 pairs $\times$ 6 strata $\times$ 3 seeds, (1, 1)).

Format axis (frozen prediction: wrapped $\geq$ raw, all 6 pair $\times$ seed conditions). Held in 5/6; one condition (code+safety, seed 42) inverts by 0.8% (16.36 raw vs. 16.23 wrapped)—reported as a falsification of the letter. The substantive comparison is one-sided everywhere: bias ratios wrapped/raw span 0.99–1.34 while the expressed-interference ratio on the same prompts is ~ 0.05 (a 20 $\times$ collapse). Representation bias does not track expression in any condition.

Pair axis (frozen two-sided). Bias orders the pairs as the output does: $\text{bias}(\text{code+safety})/\text{bias}(\text{code+math}) = 2.83/3.05/2.96$ across seeds on code prompts. This places representation bias in the same family as cumulative $\|I\|$ (§7): a coarse pair-level magnitude signal—expected, since it contains the amplified cumulative term—with no visibility into expression gating.

I BEHAVIORAL ENDPOINTS: THREE ATTEMPTS, REPORTED IN FULL

Attempt 1–2 (discrete scoring, null instruments). Unit-test pass rate on frozen code tasks and reference-answer containment on math, scored on prefill-patched greedy generations at depths 20 and 25 (2 pairs $\times$ 3 seeds). At 1.5B every code condition scored 0.0 (the code expert itself: 0.03/0.08/0.00) and math saturated at ceiling with the merge *above* its experts. At 7B the pattern replicated: code expert 0.000 against merge 0.083—the instrument does not separate models, so no verdict about the mechanism is possible from it. We report both as null instruments, not as evidence.

Attempt 3 (continuous NLL, valid instrument, negative result). Endpoint: the merged model’s mean teacher-forced NLL of a frozen 48-token continuation generated once by the task expert—continuous, threshold-free, discriminative on raw prompts, and independent of the additive counterfactual. A preregistered validity gate required the instrument to see

merge degradation at all; it passed 3/3 (merge—expert $+0.0256/+0.0218/+0.0114$; base further at $+0.09$). The frozen predictions then failed: D -style erasure at depths 20/25 changed the endpoint by $-0.00003/+0.00012/+0.00212$ (recovery share $+0.1\%/-0.6\%/-18.5\%$), and 11 of 12 control contrasts went the wrong way, with the wrong-pair control giving the *lowest* NLL in one seed. Post hoc readings we flag as untested: the effect being sought is the behavioral projection of a $\sim 19\%$ reduction in expressed non-additivity; expert-likeness is not task correctness; and merge degradation need not lie along the expert direction. The bounded conclusion stands: erasure controls expressed non-additivity, not behavioral expert-likeness.

J MULTI-WRAPPER GENERALIZATION OF THE GATE

Frozen predictions before any data: three semantically equivalent instruction templates (Alpaca plus two new phrasings) and a token-length-matched non-instruction prefix control, on both pairs $\times 3$ seeds $\times 60$ prompts, (1, 1). Result: every instruction template inflates the main-effect denominator $11\text{--}35\times$ over raw and reduces R (ratios $0.09\text{--}0.60$) in 18/18 conditions—the frozen “ $R \leq \text{raw}/2$ ” letter held in 17/18, missing once at 0.60, reported as the criterion requires. The length control *falsified* our frozen expectation that its denominator would stay within $2\times$ (observed $3.5\text{--}6.7\times$): long prefixes do shift the distribution. The discriminating quantity the data identify is R itself: the non-instruction prefix inflates numerator and denominator together (numerator $3.4\text{--}8.3\times$), leaving R directionally unchanged ($0.59\text{--}2.4$), whereas instruction templates pin the distribution faster than they excite the interaction. Across all wrappers, including the control, the *absolute* interaction numerator increases ($1.8\text{--}11\times$)—the fifth independent replication of internal amplification under wrapping.

K SCALE POINT (QWEN2.5-7B)

Same family, larger scale, as the third grid point; 28 blocks, so the frozen boundaries carry over unchanged. Nine adapters with the identical recipe; reduced KT1 (54 cells), amplification bounds, and the *full* 54-run KT2 in float32 with per-prompt persistence (run on a single A800-80GB, the same GA100 die as the A100; the card-type amendment was logged before any KT2 data existed). Four predictions frozen before any 7B data.

Band and output mismatch (frozen gate missed by 0.1%). The G -band ratio across the 18 (1, 1) cells is 1.301 against the frozen <1.3 gate—a falsification by the letter, reported as such. The substantive contrast sharpens with scale: output pair ratios $6.34/12.20/7.19$ (vs. $2\times$ at 1.5B) and format ratios $6.45/7.54/6.33$, against a band spread of ≤ 1.30 ; and $G_{\text{wrapped}} > G_{\text{raw}}$ replicates 3/3—the third model point with the reversal.

Amplification. $\|I\|$ exceeds the coherent bound by $1.80\text{--}8.31\times$ across conditions (frozen gate >1.2 , 3/3). Super-coherent amplification is again largest under the wrapper in both pairs, and the raw—wrapped late-gain contrast is $+0.091/+0.091/+0.073$ —positive 3/3, the largest of the three model points. The pair-axis late-gain contrast is not seed-consistent (exploratory, reported).

Causal triple (all three legs, all bootstrap intervals excluding zero). $D_{\text{correct}} = +0.0274/+0.0549/+0.0270$ with all 15 specificity intervals excluding zero. Expression gating is *total* at this scale: $D_{\text{wrapped}} = +0.0040/-0.0038/+0.0007$, each CI containing zero, while $D_{\text{raw}} - D_{\text{wrapped}} = +0.0658/+0.0596/+0.0475$, all excluding zero—under the wrapper there is nothing left to erase. Pair contrasts $+0.0241/+0.0525/+0.0231$ all exclude zero, implying removable-interference ratios $8.3/22.9/6.9$ between pairs whose generation magnitudes differ by $\leq 1.3\times$.

L EXACT TRANSPORT DECOMPOSITION AND THE FP32 GENERATION GRID

Both experiments were preregistered together (five predictions with priors, frozen before any decomposition or fp32-grid data existed) as the data-level adjudication of two review-critical questions: whether the ledger’s remainder can be attributed to transport, and whether the bf16 uniformity band survives at float32. Scope: 36 cells (2 pairs $\times 6$ strata $\times 3$ seeds, (1, 1), all prompts, all 28 layers); the model is loaded in bf16 (the measurement state of the main sweep) and each block is evaluated with float32 weights and captured call arguments. The regrouping identity $I_{\ell+1} = G_{\ell} + T_{\ell} + M_{\ell}$ is asserted numerically at every layer of every cell.

E4 (transport): both frozen predictions held; the two-sided share resolved to transport-dominant. Late-network $\text{RMS}(T_{\ell})$ exceeds $\text{RMS}(G_{\ell} + M_{\ell})$ by $2.3\times$ (0.32 vs. 0.14 , 3/3 seeds); the transport gain $\|T_{\ell}\|/\|I_{\ell}\|$ averages

1.078–1.079 over the late half (3/3); the regeneration share of the total flux is 0.307–0.311 (3/3), so transport carries $\sim 69\%$ under the norm accounting. Because the split is non-orthogonal in principle, we also computed the components’ alignment and an energy (symmetric-projection) accounting on the code cells: late-network $\cos(T_\ell, G_\ell + M_\ell) = -0.014$ to -0.031 (6/6 near-orthogonal), and the energy-accounting transport share is 0.83–0.89—the attribution is essentially accounting-independent. Combined with the E1 regrowth result—regeneration alone suffices to rebuild the full term, in norm and direction, given ≥ 14 blocks—the consistent picture is a transport-dominated flux whose fixed direction the regeneration channel actively restores: an attractor.

E5 (fp32 grid): both frozen predictions falsified, and reported as the criteria require. The fp32 $\text{AUC}(G)$ band across the 36 cells is $3.36 \times (0.0012\text{--}0.0040)$; frozen gate < 2 , and the pair ratio is $1.81\text{--}1.90 \times$ (frozen gate < 1.5): float32 local generation is *not* uniform, and it coarsely tracks the pair ordering—joining cumulative norm, parameter cosine, and representation bias in the coarse-magnitude family. The format axis preserves the dissociation: fp32 generation is $1.66\text{--}1.84 \times$ *larger* under the wrapper (raw/wrapped ratios 0.54–0.60, 3/3 seeds) while expressed interference collapses $\sim 20 \times$. The bf16 $\pm 15\%$ band of §4 is therefore confirmed to be floor-compression—the cautionary tale stands in its strongest form, and the magnitude-fallacy claim rests on the causal contrast and the format inversion, not on band uniformity.

Llama replication (frozen before its data): transport invariant, magnitude behavior not. On Llama-3.2-1B (36 cells, 16 layers) both transport predictions held again—late-network $\text{RMS}(T)$ exceeds $\text{RMS}(G + M)$ by $\sim 2 \times$, transport gain 1.14 per late block, regeneration share 0.35 (all 3/3). The fp32 magnitude gates, by contrast, resolved *opposite* to Qwen—band ratio 1.65 (within the < 2 gate), pair ratio 1.06 (no pair separation at all, against an output ratio of $\sim 1.5 \times$)—falsifying our frozen expectation that the Qwen pattern would replicate isomorphically. The cross-family summary is sharper than either result alone: transport dominance and the format inversion ($G_{\text{wrapped}} > G_{\text{raw}}$; raw/wrapped 0.54–0.60 on Qwen, 0.78–0.84 on Llama, each 3/3) are family invariants, while whether local-generation magnitude tracks pairs at all is itself a family property—magnitude is not merely a coarse signal, it is an *inconsistent* one.